

Project 2: Cohort Retention & Customer Lifetime Value (CLTV) Analysis

Executive Business Insights Report

1. Executive Summary

This report summarizes a deep-dive cohort retention and customer lifetime value (CLTV) analysis performed on 2,211 completed transactions from 1,546 paying customers, spanning January 2023 to June 2024. The objective was to identify when and why customers churn, and to quantify the lifetime value of different customer segments in order to guide retention strategy and marketing budget allocation.

Headline Metrics

₹70.7L Total Revenue	1,546 Total Customers	₹3,227 Avg. Order Value	₹4,580 Avg. CLTV
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2. Exploratory Data Analysis — Visual Overview

Before building the cohort and CLTV models, the cleaned transaction data was explored visually to understand its underlying structure across time, region, and acquisition channel.

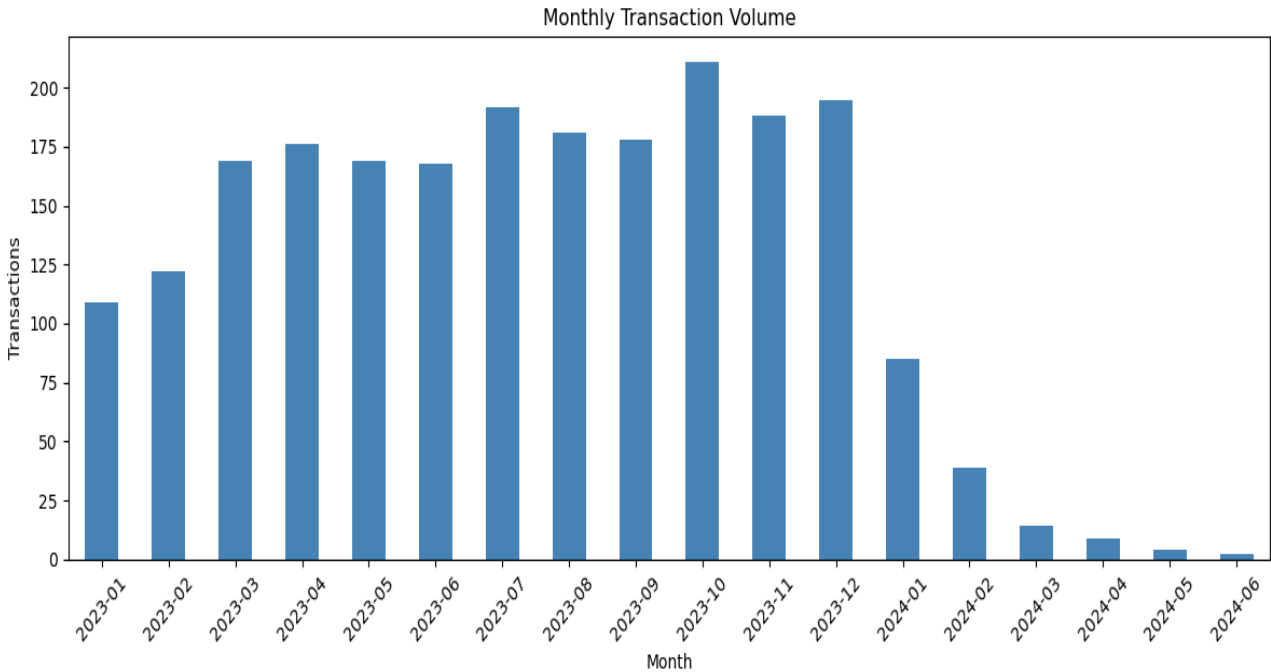


Figure 1: Monthly Transaction Volume (Jan 2023 – Jun 2024)

Transaction volume grew steadily through 2023, peaking in October 2023, before tapering off sharply from early 2024 onward — consistent with the dataset's observation window closing and newer cohorts not yet having time to accumulate repeat purchases.

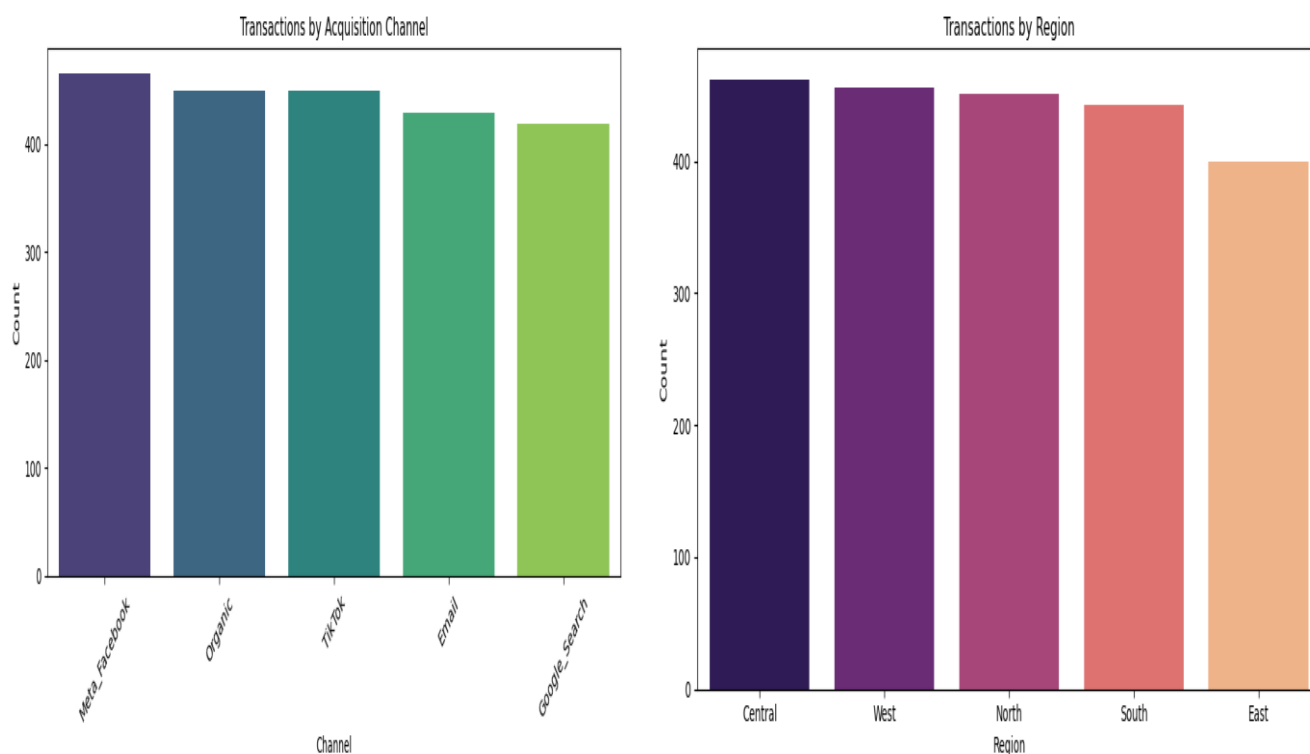


Figure 2: Transactions by Acquisition Channel (left) and by Region (right)

Transaction volume is fairly evenly distributed across all five acquisition channels and all five regions, with no single source dominating — meaning differences in CLTV across segments (covered in Section 4) reflect genuine differences in customer value rather than sample-size skew.

3. Cohort Retention Analysis

Customers were grouped into monthly cohorts based on their first transaction date. Each cohort's activity was then tracked over subsequent months to calculate the percentage of users retained at Month 0, Month 1, Month 2, and so on.

Key Finding: Steep First-Month Drop-off

- Month 0 retention is 100% by definition (the acquisition month itself).
- Retention falls to roughly 14%–20% by Month 1 across nearly all cohorts — the single largest drop in the customer lifecycle.
- From Month 2 onward, retention decays more gradually, typically settling in the low single digits by Month 4–5.
- This pattern is consistent across cohorts from Jan 2023 through Dec 2023, indicating a structural, product-wide retention issue rather than a one-off seasonal effect.

The cohort retention heatmap below visualizes this decay pattern across all 16 monthly cohorts and up to 7 months of follow-up. Darker cells indicate higher retention; the sharp colour drop from Month 0 to Month 1 across every row confirms this is a consistent, cohort-independent pattern.

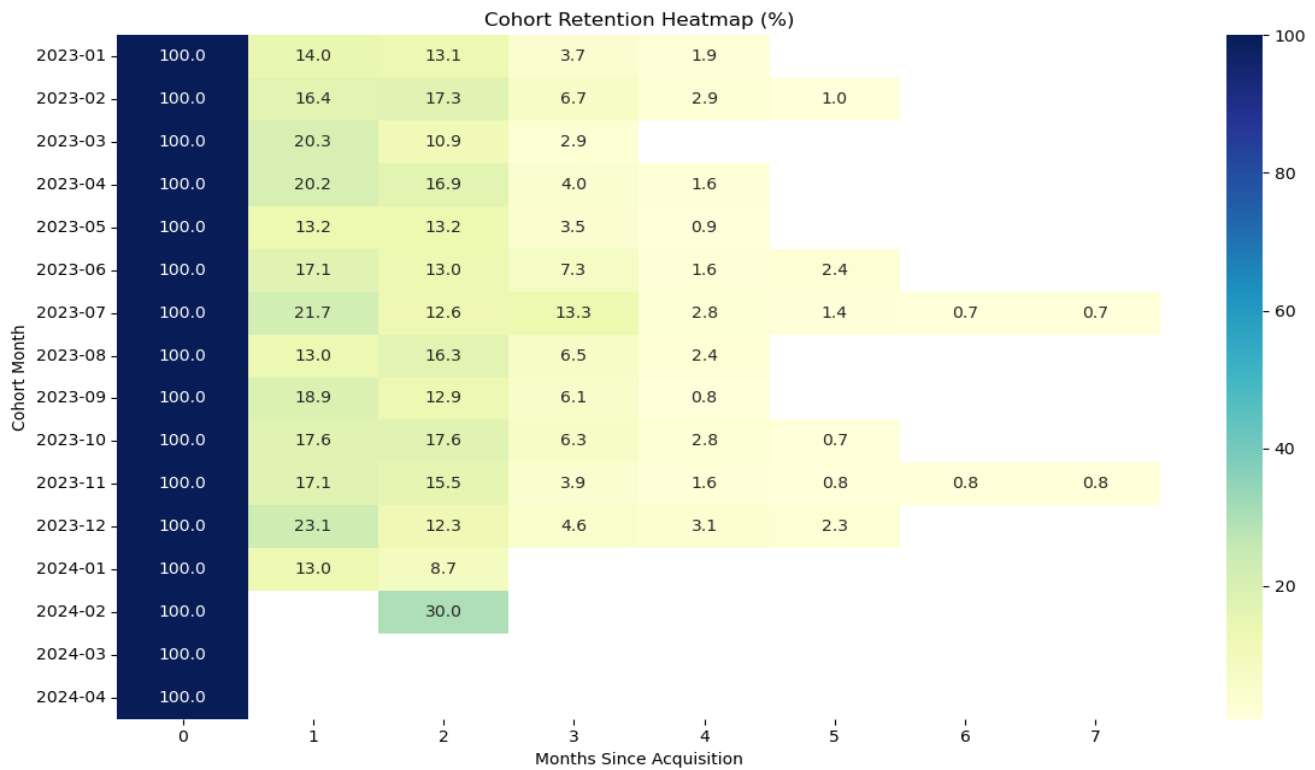


Figure 3: Cohort Retention Heatmap (% retained by cohort month and months since acquisition)

4. Customer Lifetime Value (CLTV)

Historical CLTV was calculated per customer as Average Order Value $\times$ Purchase Frequency, using each customer's actual completed transaction history. This was then aggregated to compare value across geographic regions and acquisition channels.

4.1 CLTV by Region

Region	Total Revenue	Customers	Avg. CLTV
South	₹15,10,626	320	₹4,720.71
West	₹14,55,988	322	₹4,521.70
North	₹14,35,423	314	₹4,571.41
Central	₹14,48,962	321	₹4,513.90
East	₹12,23,673	269	₹4,548.97

- South leads on both total revenue and average CLTV, making it the strongest-performing region overall.
- East has the lowest customer count (269) and the lowest total revenue, signalling an acquisition gap rather than a value gap — worth investigating for under-investment in marketing.

4.2 CLTV by Acquisition Channel

Channel	Total Revenue	Customers	Avg. CLTV
TikTok	₹15,18,284	310	₹4,897.69
Meta Facebook	₹15,08,135	324	₹4,654.74
Google Search	₹13,47,951	299	₹4,508.20
Organic	₹13,58,151	310	₹4,381.13
Email	₹13,42,151	303	₹4,429.54

- TikTok-acquired customers carry the highest average CLTV (₹4,897.69), followed closely by Meta Facebook — both paid social channels outperform organic and email on customer value.
- Organic traffic, despite having no acquisition cost, produces the lowest average CLTV — volume without proportional long-term value.
- This suggests marketing spend on TikTok and Meta is currently justified by downstream customer value, not just top-of-funnel volume.

4.3 Top Customers

The ten highest-CLTV customers are led by a single customer (USR_01998) contributing approximately ₹12,200 in lifetime value — over 2.6x the average. High-CLTV customers disproportionately drive revenue and are strong candidates for a dedicated loyalty or VIP retention program.

5. Recommendations

- Launch an automated re-engagement email sequence targeting customers around the Month-1 mark, when the steepest retention drop occurs.
 - Reallocate a larger share of acquisition budget toward TikTok and Meta Facebook, given their superior average CLTV relative to Organic and Email.
 - Investigate under-acquisition in the East region — its per-customer value is comparable to other regions, but customer volume lags, pointing to a marketing reach gap rather than a product-fit issue.
 - Build a VIP program for the top-decile CLTV customers to protect and grow the revenue they already contribute disproportionately.
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6. Dashboard Preview

The interactive Power BI dashboard below allows stakeholders to filter by Region and Acquisition Channel, and surfaces Total Revenue, Total Customers, Average Order Value, Average CLTV, the Top 10 customers by CLTV, and revenue distribution across regions and channels.

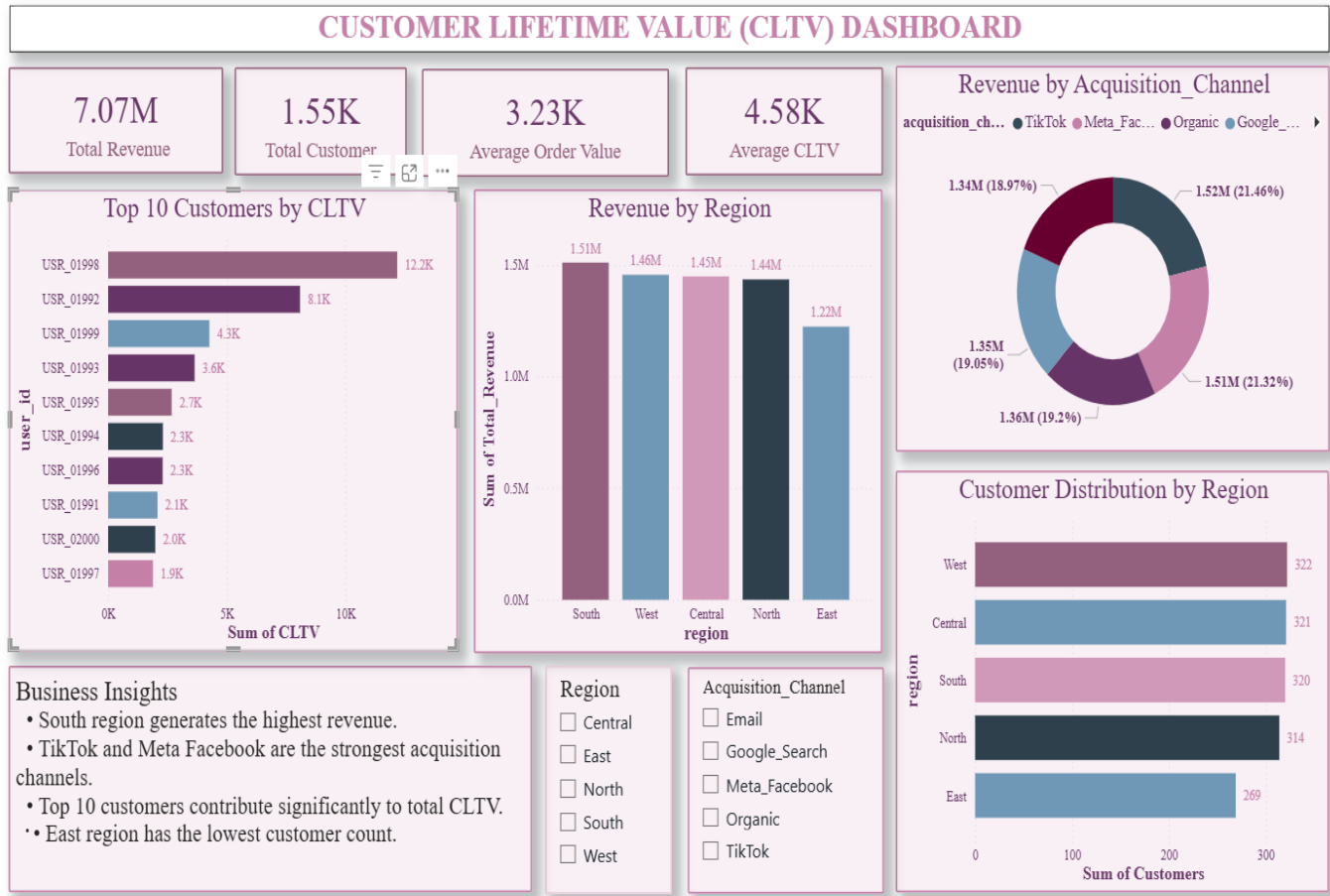


Figure 4: CLTV Dashboard — Power BI (CLTV_Dashboard.pbix)

7. Deliverables

- 01_EDA_Cleaning.ipynb — data cleaning and cohort-month assignment
- 02_Cohort_Matrix.ipynb — cohort retention matrix and heatmap
- 03_CLTV_Calculation.ipynb — AOV, purchase frequency, historical and segment-wise CLTV
- CLTV_Dashboard.pbix — interactive Power BI dashboard
- Processed CSVs — cohort_pivot, retention_matrix, customer_cltv, region_summary, channel_summary, segment_cltv_region, segment_cltv_channel